

AI Job Agent: System Architecture & Workflow Blueprint

End-to-End Autonomous Job Discovery, AI Scoring, Dynamic ATS Tailoring & Multi-Mode Application System

Target Candidate: Kamal Kumar	Primary Tech Stack: Java, Spring Boot, Python, FastAPI, AWS, Docker, AI
Experience Level: 0 to 2 Years ONLY (Junior / Entry-Level)	Active Connectors: 16 Sources across APIs, ATS, & Social Posts
Execution Cycle: 24/7 Fixed Runs (06:00, 12:00, 18:00, 00:00 IST)	Core Framework: LangGraph + FastAPI + Playwright + Gemini AI

1. Executive Overview & Architecture

The **AI Job Agent** is an enterprise-grade, autonomous job application ecosystem designed to solve the challenges of modern technical hiring. Unlike generic scrapers or basic autofill bots, this system provides a **resilient, 6-stage LangGraph workflow** powered by Google Gemini AI, strict heuristic experience filters, dynamic 1-page ATS PDF generation, and a Source-Aware 3-Mode Application Engine.

Core Architectural Pillars:

- Multi-Channel Ingestion:** Concurrently queries 16 diverse job sources including enterprise ATS endpoints (Greenhouse, Lever, Ashby, Workday), public remote job APIs (Jobicy, Remotev, Himalayas, Adzuna), and social recruiter posts (LinkedIn, HackerNews).
- Zero-Tolerance Experience Guard:** A multi-layer heuristic and semantic filter that automatically hard-blocks any role requiring >2 years of experience or containing senior/lead keywords before expensive LLM processing.
- 100% Unique Dynamic ATS Resumes:** For every qualifying role, Gemini extracts required skills and dynamically compiles a bespoke single-page PDF resume using ReportLab Platypus.
- Source-Aware 3-Mode Application Policy:** Distinguishes between programmatic ATS APIs, persistent browser automation, verified HR email outreach, and 1-click manual links.
- Real-Time Multi-Channel Telemetry:** Delivers instant daily summaries and high-match alerts via WhatsApp (CallMeBot / Meta Cloud API) and styled multi-tab Excel workbooks.

2. 6-Stage LangGraph Pipeline Workflow

The pipeline is implemented as a deterministic **LangGraph StateGraph**. The agent state passes sequentially through 6 specialized nodes:

Stage & Node	Primary Responsibilities	Outputs & Artifacts
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Node 1: Ingestion fetch_node	• Concurrently polls 16 job connectors. • Extracts title, company, location, JD, and URL. • SQLite SHA-256 deduplication against past runs.	• raw_jobs list • new_jobs_count • Saved in SQLite DB
Node 2: AI Scoring score_node	• Stage 1: Heuristic pre-scoring & 0-2 yrs hard guard. • Stage 2: Gemini AI semantic scoring (0-100). • Evaluates skill overlap, experience fit, and role match.	• scored_jobs list • qualified_job_ids • Match score (0-100)
Node 3: ATS Resume resume_node	• Extracts tech keywords from target job description. • Gemini tailors summary, skill badges, and bullet points. • Builds ATS single-page PDF via ReportLab Platypus.	• resume_paths map • Tailored PDF in resumes/
Node 4: Apply Engine apply_node	• Evaluates Source-Aware 3-Mode policy. • Executes Mode 1 API, Mode 2 Persistent Browser, or Mode 3 HR Email. • Prepares 1-click link for anti-bot aggregator portals.	• direct_applied count • emails_sent count • DB status: APPLIED
Node 5: WhatsApp notify_node	• Formats clean daily WhatsApp summary. • Sends top-3 job match alerts with scores & links. • Supports CallMeBot free instant API & Meta Cloud API.	• WhatsApp message delivered • whatsapp_sent=True
Node 6: Reporting report_node	• Compiles multi-tab Excel workbook via openpyxl. • Tabs: <i>All Discovered</i>, <i>Qualified Matches</i>, <i>Applications</i>. • Saves daily report with clickable direct URLs.	• JobReport_YYYY-MM-DD.xlsx • Stored in reports/

3. 16 Active Job Connectors & Ingestion Layer

The agent integrates with 16 distinct connectors managed by a centralized ConnectorManager. Connectors execute asynchronously with error isolation — if a single third-party endpoint fails or rate-limits, the remaining 15 connectors continue uninterrupted.

Connector Name	Source / Portal	Type & Protocol	Target Domain
GreenhouseConnector	Greenhouse Enterprise ATS	Public REST API (JSON)	Airbnb, Stripe, Brex, Discord, Figma
LeverConnector	Lever Enterprise ATS	Public REST API (JSON)	Netflix, Spotify, Wealthsimple
AshbyConnector	Ashby Modern ATS	Posting REST API	Linear, Supabase, Retool, Vercel
WorkdayConnector	Workday CXS Portal	CXS Search API / HTTP	Walmart, Target, Flipkart, Adobe
SmartRecruitersConnector	SmartRecruiters ATS	REST API	Freshworks, Visa, Bosch
JobicyConnector	Jobicy Remote Jobs	Public Jobs API	Global Remote Software & AI Roles
RemotiveConnector	Remotive Software API	Public Category API	Backend, Cloud & AI Engineering
HimalayasConnector	Himalayas Remote	Public REST API	Modern Tech Startups & Scaleups
AdzunaConnector	Adzuna India & Global	Developer REST API	India Tech Market (Java/Python/AI)
AgentReachConnector	Agent Reach LinkedIn	CLI / Headless Protocol	LinkedIn Posts & Direct Roles
LinkedInPostsConnector	LinkedIn Hiring Posts	Direct HR Post Scraper	Recruiter Cold Hiring Posts
YCombinatorConnector	WorkAtAStartup / YC	HackerNews Algolia API	YC Backed High-Growth Startups
WellfoundConnector	Wellfound (AngelList)	Public RSS / Discovery	Early-Stage & Seed AI Startups
ArbeitsnowConnector	Arbeitsnow Tech	REST API	European & Remote Tech Roles
WeWorkRemotelyConnector	We Work Remotely	RSS & JSON Feed	Remote Backend & Systems Roles
RemoteOKConnector	RemoteOK	Public REST API	Global Remote Tech & Fullstack

4. Source-Aware Application Policy & Experience Guard

The agent implements a **Source-Aware Multi-Mode Application Architecture** to interact appropriately with each portal type:

Mode	Strategy & Channel	Supported Platforms	Behavior & Automation Logic
■ Mode 1 Direct API	Programmatic ATS Form Submission	Greenhouse, Lever, Ashby	Submits multipart/form-data with candidate profile, questions, and tailored ATS resume PDF directly via official API.
■ Mode 2 Persistent Browser	Playwright Persistent Context	Wellfound, YCombinator, LinkedIn, Workday, Jobicy	Reuses authenticated local browser profile (data/browser_profile). Auto-fills forms, uploads resume PDF, and auto-submits without password leakage.
📧 ■ Mode 3 Verified HR Email	Direct Recruiter Outreach via Gmail API	LinkedIn Posts, HackerNews hiring threads	Zero Bounce Policy: Sends cold application ONLY when an explicit recruiter email is parsed from the post. Never guesses synthetic addresses.

■ Mode 4 1-Click Link Prep	Custom Resume & Direct Link Preparation	Adzuna, Hirst, Shine, Cutshort, Foundit	Generates custom ATS resume PDF, drafts cover note, and formats 1-click direct application link in the daily Excel report.
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Strict Match Score Policy:

- **Score ≥ 85:** AUTO_APPLY — Programmatic submission via Mode 1 (API) or Mode 2 (Persistent Browser).
- **Score 75–84:** REVIEW_REQUIRED — Custom ATS resume PDF generated + draft saved for 1-click submission.
- **Score 65–74:** SAVE_LINK — Job details, cover letter draft, and direct URL saved in Excel report.
- **Score < 65:** SKIP — Discarded from scoring queue.

5. Complete Project Directory & Codebase Structure

Below is the complete file tree and architectural module map of the AI-Job-Agent codebase:

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AI-Job-Agent/ ■■■ app/ ■■■ agents/ # LangGraph StateGraph & Node Orchestrators ■ ■ ■■■
__init__.py # Agent package exports ■ ■ ■■■ nodes.py # 6 Pipeline Nodes (fetch, score, resume, apply,
notify, report) ■ ■ ■■■ orchestrator.py # LangGraph StateGraph compilation & execution graph ■ ■
■■■ state.py # AgentState TypedDict schema definition ■ ■■■ config/ # Centralized Configuration &
Environment Settings ■ ■ ■■■ __init__.py # Config exports ■ ■ ■■■ settings.py # Environment
variables, candidate profile, score thresholds ■ ■■■ connectors/ # 16 Job Search Ingestion Connectors
■ ■ ■■■ __init__.py # Connector package registry ■ ■ ■■■ base.py # BaseConnector abstract class
interface ■ ■ ■■■ manager.py # ConnectorManager concurrent aggregator ■ ■ ■■■ registry.py #
Connector auto-discovery decorator ■ ■ ■■■ greenhouse_connector.py # Greenhouse ATS REST API
connector ■ ■ ■■■ lever_connector.py # Lever ATS REST API connector ■ ■ ■■■ ashby_connector.py #
Ashby ATS REST API connector ■ ■ ■■■ workday_connector.py # Workday CXS Search API connector ■ ■
■■■ smartrecruiters_connector.py # SmartRecruiters ATS connector ■ ■ ■■■ jobcity_connector.py #
Jobcity Remote Jobs API connector ■ ■ ■■■ remotive_connector.py # Remotive Software Jobs API connector
■ ■ ■■■ himalayas_connector.py # Himalayas Remote API connector ■ ■ ■■■ adzuna_connector.py #
Adzuna Developer Jobs API connector ■ ■ ■■■ agent_reach_connector.py # Agent Reach LinkedIn scraper
■ ■ ■■■ linkedin_posts_connector.py # LinkedIn Recruiter Hiring Posts parser ■ ■ ■■■
ycombinator_connector.py # HackerNews / YC Who is Hiring Algolia API ■ ■ ■■■ wellfound_connector.py #
Wellfound (AngelList Talent) connector ■ ■ ■■■ arbeitnow_connector.py # Arbeitnow Tech Jobs API
connector ■ ■ ■■■ weworkremotely_connector.py # WeWorkRemotely RSS feed connector ■ ■■■ db/ #
Database Storage & Repository Layer ■ ■ ■■■ __init__.py # DB models & session exports ■ ■ ■■■
models.py # SQLAlchemy ORM models (Job, Application, Resume, RunLog) ■ ■ ■■■ session.py # SQLite
engine & scoped session factory (jobs.db) ■ ■■■ prompts/ # LLM Prompt Templates & System Personas ■
■■ ■■■ __init__.py # Prompts package ■ ■ ■■■ email_prompt.py # Cold outreach email generation prompt
■ ■ ■■■ resume_prompt.py # Dynamic ATS resume tailoring prompt ■ ■ ■■■ scoring_prompt.py # Strict
0-2 yrs technical job scoring prompt ■ ■■■ schemas/ # Pydantic Schemas & Data Transfer Objects ■ ■
■■■ __init__.py # Schemas package ■ ■ ■■■ application.py # Application status & tracking schema ■ ■
■■■ job_data.py # Unified JobData dataclass ■ ■■■ services/ # Core Business Logic & AI Services ■ ■
■■■ __init__.py # Services package exports ■ ■ ■■■ application_policy.py # Source-Aware 3-Mode
Application Strategy Manager ■ ■ ■■■ apply_service.py # Application submission cascade & routing
service ■ ■ ■■■ browser_apply_service.py # Playwright Chromium persistent browser auto-apply ■ ■
■■■ email_service.py # Gmail API OAuth2 cold outreach sender ■ ■ ■■■ mock_scoring_service.py #
Heuristic pre-filter & keyword scoring engine ■ ■ ■■■ report_service.py # openpyxl multi-tab Excel
report generator ■ ■ ■■■ resume_service.py # Dynamic ATS 1-page PDF builder (ReportLab Platypus) ■ ■
■■■ scheduler_service.py # APScheduler 24/7 fixed 6-hour cron service ■ ■ ■■■ scoring_service.py #
Google Gemini AI LLM semantic scoring service ■ ■ ■■■ whatsapp_service.py # CallMeBot & Meta Cloud
WhatsApp alert service ■ ■■■ utils/ # Shared Helper Functions & Validators ■ ■■■ __init__.py # Utils
package ■ ■■■ email_validator.py # Explicit HR email regex parser & DNS validator ■ ■■■
experience_filter.py # Strict 0-2 years experience & senior title hard-guard ■ ■■■ gemini_client.py #
Resilient Gemini API client with auto-retry ■ ■■■ data/ # Persistent Storage & Browser Profiles ■ ■■■
browser_profile/ # Playwright persistent Chromium context & cookies ■ ■■■ jobs.db # SQLite primary
database ■ ■■■ docs/ # Architecture Documentation & PDF Blueprints ■ ■■■ reports/ # Daily Excel
Workbooks (JobReport_YYYY-MM-DD.xlsx) ■ ■■■ resumes/ # Master Resume & Tailored ATS PDF Resumes ■ ■■■
Kamal_Kumar_Java_AI_Developer_ATS.pdf # Candidate Master Resume ■ ■■■ scripts/ # Diagnostic, Testing, &
Execution Tools ■ ■■■ run_agent.py # CLI runner for single pipeline execution ■ ■■■
test_browser_apply.py # Live visible browser auto-apply tester ■ ■■■ setup_gmail_token.py # Gmail
OAuth2 token generator ■ ■■■ setup_whatsapp.py # WhatsApp CallMeBot diagnostic tool ■ ■■■
generate_architecture_pdf.py # Architecture blueprint PDF generator ■ ■■■ main.py # FastAPI 24/7 Web
Server & Background Scheduler Entrypoint ■ ■■■ requirements.txt # Production Python dependencies ■ ■■■
.env # Configuration & API Credentials

```

6. Server Lifecycle, Scheduling, & API Endpoints

The application runs 24/7 as a **FastAPI service** with an embedded **APScheduler** background engine. It provides complete observability through REST API endpoints and automatic periodic executions.

HTTP Method & Endpoint	Description & Functionality	Sample Response / Action
GET /	System health check, version, and scheduler status.	{'status': 'running', 'scheduler': True}
POST /agent/run	Triggers an immediate background run across all 16 connectors.	{'status': 'started', 'run_id': '41fee100'}
GET /agent/status	Retrieves the execution status and statistics of the latest run.	{'last_run_time': '...', 'jobs_found': 3539}
GET /jobs/qualified	Lists all qualified matches (score ≥ 65) matching 0-2 yrs target.	[{'title': 'Java Dev', 'score': 85}, ...]
GET /jobs/stats	Summary statistics of discovered, scored, and applied jobs.	{'total_jobs': 3539, 'applied': 12}
GET /docs	Interactive Swagger UI for testing API endpoints.	Interactive Web Interface

7. Architectural Summary & Production Readiness

With this architecture, the **AI Job Agent** functions as an autonomous career assistant that discovers thousands of opportunities daily, filters with strict 0–2 years precision, tailors ATS resumes on the fly, applies across multiple channels without password exposure, and keeps the candidate informed 24/7.

■ Multi-Source Discovery: 16 integrated sources querying 3,500+ jobs per cycle.	■ 0-2 Years Precision: Hard guards eliminate all >2 yrs & senior roles.
■ Zero Email Bounces: Strictly emails explicit recruiter contacts from posts.	■ 100% Unique Resumes: Tailored ATS single-page PDFs per application.
■ Safe Browser Sessions: Persistent cookies reuse without password leaks.	■ Daily Multi-Channel Alerts: Automated WhatsApp alerts & Excel reports.